Separable Strongly Rosenthal Compacta and the Admissible Envelope Reduction

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Status

This packet records a likely valid partial result for the question of Glasner and Megrelishvili [1]. It proves the full separable subcase:

Every separable strongly Rosenthal compactum is admissible.

It also gives the following exact reformulation:

K is admissible if and only if K is a compact subspace of a separable strongly Rosenthal compactum.

Consequently the full source question is equivalent, after this packet, to the separable-envelope problem: does every strongly Rosenthal compactum embed into a separable strongly Rosenthal compactum? The nonseparable case remains open here.

Source Question

The source paper defines a compactum K to be strongly Rosenthal when it embeds into $\mathcal{B}_1(X)$ for a compact metrizable space X, and admissible when it is contained in the pointwise closure of a bounded set of continuous functions whose pointwise closure consists of Baire-one functions. The source then asks whether the two classes coincide.

continuity property when restricted to B^*. Equivalently, every restriction of v^{**} to a bounded subset M is fragmented as a function $(M, w^*) \to \mathbb{R}$ (see Fact 4.11 below).

Answering a question of Talagrand [42, Problem14-2-41], R. Pol [36] gave an example of a separable compact Rosenthal space K which cannot be embedded in $\mathcal{B}_1(X)$ for any compact metrizable X. We say that a compact space K is *strongly Rosenthal* if it is homeomorphic to a subspace of $\mathcal{B}_1(X)$ for a compact metrizable X. We say that a compact space K is *admissible* if there exists a metrizable compact space X and a bounded subset $Z \subset C(X)$ such that the pointwise closure $\mathrm{cls}_p(Z)$ of Z in $\mathbb{R}^X$ consists of Baire 1 functions and $K \subset \mathrm{cls}_p(Z)$. Clearly every admissible compactum is strongly Rosenthal. We do not know whether these two classes of compact spaces coincide.

Definitions and Source Tools

Let X be compact metrizable. A bounded family $F \subset C(X)$ is a Rosenthal family if $\mathrm{cls}_p(F) \subset \mathcal{B}_1(X)$. In the source paper this is equivalent, by the Talagrand/Rosenthal criterion, to saying that F contains no sequence equivalent to the unit vector basis of ℓ_1 [1].

A compactum is admissible if it is homeomorphic to a compact subspace of $\mathrm{cls}_p(F)$ for some Rosenthal family $F \subset C(X)$ on a compact metrizable X.

We use two results from Glasner–Megrelishvili [1].

- Theorem 6.3: if $F \subset C(Y)$ is a Rosenthal family on compact metrizable Y, then $L = \mathrm{cls}_p(F)$ embeds into V^{**}, with the weak-star topology, for a separable Rosenthal Banach space V. Moreover, for the trivial action version used here, the embedding extends the original functions in F and represents evaluation on Y by functionals in V^*.
- Theorem 6.16: a compactum is admissible if and only if it is homeomorphic to a weak-star closed subset of V^{**} for some separable Rosenthal Banach space V.

Proof Intuition

The previous push produced an admissible cover. Given a separable compact $K \subset \mathcal{B}_1(X)$, choose a dense sequence (f_n) in K, compactify the graph

$$x \mapsto (x, f_0(x), f_1(x), \ldots),$$

and let g_n be the coordinate functions on the compactified graph. Then (g_n) is a Rosenthal family of continuous functions, and $L = \mathrm{cls}_p(\{g_n\})$ maps continuously onto K by restriction to the original graph.

The missing step was not to find a continuous section of this quotient. The new idea is to quotient the Banach space representing L. The source representation theorem embeds L into V^{**} for a separable Rosenthal space V. Let $N \subset V$ be the closed subspace of vectors whose represented functions vanish on the original graph points. Passing from V to $W = V/N$ collapses precisely those elements of L that have the same restriction to the original graph. Therefore the quotient bidual W^{**} contains a weak-star compact copy of K. The source second-dual characterization then implies that K is admissible.

Theorem

Theorem 1. *Every separable strongly Rosenthal compactum is admissible.*

Proof. Since the property is topological, compose with a homeomorphism $\mathbb{R} \to (-1, 1)$ coordinatewise if necessary. Thus it is enough to consider a compact separable set

$$K \subset [-1, 1]^X \cap \mathcal{B}_1(X),$$

where X is compact metrizable and K has the pointwise topology. Choose a countable pointwise-dense sequence $(f_n)_{n\in\mathbb{N}}$ in K.

Define

$$e : X \to X \times [-1, 1]^{\mathbb{N}}, \qquad e(x) = (x, f_0(x), f_1(x), \ldots),$$

and let

$$Y = \overline{e(X)} \subset X \times [-1,1]^{\mathbb{N}}.$$

The space Y is compact metrizable. Let $g_n : Y \to [-1,1]$ be the n-th coordinate function, so $g_n \in C(Y)$ and

$$g_n(e(x)) = f_n(x) \qquad (x \in X).$$

First we recall the cover argument. The family $F = \{g_n : n \in \mathbb{N}\}$ is a Rosenthal family on Y. Indeed, if some subsequence (g_{n_j}) were equivalent to the unit vector basis of ℓ_1, then by the Talagrand/Rosenthal finite-independence criterion there would be real numbers $a < b$ such that for every finite disjoint $P, Q \subset \mathbb{N}$ the inequalities

$$g_{n_j}(y) < a \quad (j \in P), \qquad g_{n_j}(y) > b \quad (j \in Q)$$

hold at some $y \in Y$. For fixed P, Q, these strict inequalities define a nonempty open subset of Y. Since $e(X)$ is dense in Y, the same pattern is realized at some $e(x)$. Hence

$$f_{n_j}(x) < a \quad (j \in P), \qquad f_{n_j}(x) > b \quad (j \in Q).$$

Thus (f_{n_j}) contains an ℓ_1-sequence, contradicting the fact that K is pointwise compact in $\mathcal{B}_1(X)$: every sequence in K has a pointwise convergent subsequence, while an ℓ_1-sequence of bounded functions has no pointwise convergent subsequence. Therefore F is a Rosenthal family.

Set

$$L = \mathrm{cls}_p(F) \subset [-1,1]^Y.$$

Then L is admissible. Define

$$\pi : L \to K, \qquad \pi(h)(x) = h(e(x)).$$

This map is continuous for the pointwise topologies. It maps g_n to f_n, and compactness gives $\pi(L) = K$: if $f \in K$, choose a net (f_{n_i}) converging pointwise to f; a subnet of (g_{n_i}) converges pointwise on Y to some $h \in L$, and then $\pi(h) = f$.

Now apply Theorem 6.3 of [1] to the Rosenthal family $F \subset C(Y)$, with the trivial group action. We obtain a separable Rosenthal Banach space V, a weak-star homeomorphic embedding

$$\nu_0 : L \hookrightarrow V^{**},$$

and a weak-star continuous map $\alpha : Y \to V^*$ such that

$$h(y) = \langle \nu_0(h), \alpha(y) \rangle \qquad (h \in L,\ y \in Y).$$

Moreover, $\nu_0(g_n) \in V$ for every n.

Let

$$N = \bigcap_{x \in X} \ker \alpha(e(x)) \subset V,$$

and let $q : V \to W := V/N$ be the quotient map. Since quotients of separable Rosenthal Banach spaces are again separable Rosenthal, W is a separable Rosenthal space.

Identify W^* with the annihilator

$$N^{\perp} = \{\varphi \in V^* : \varphi|_N = 0\}.$$

By the elementary annihilator identity,

$$N^{\perp} = \overline{\mathrm{span}}\{\alpha(e(x)) : x \in X\} \quad \text{inside } V^*.$$

We claim that $q^{**} \circ \nu_0$ collapses exactly the fibers of π. Let $h_1, h_2 \in L$. Then

$$\begin{aligned}
\pi(h_1) = \pi(h_2) &\iff h_1(e(x)) = h_2(e(x)) \quad \forall x \in X \\
&\iff \langle \nu_0(h_1) - \nu_0(h_2), \alpha(e(x)) \rangle = 0 \quad \forall x \in X \\
&\iff \nu_0(h_1) - \nu_0(h_2) \text{ vanishes on } N^{\perp} \\
&\iff q^{**}\nu_0(h_1) = q^{**}\nu_0(h_2).
\end{aligned}$$

The third equivalence uses that elements of V^{**} are norm-continuous on V^*, so vanishing on the spanning set implies vanishing on its closed linear span.

It follows that the continuous map $q^{**} \circ \nu_0 : L \to W^{**}$ factors through the quotient map $\pi : L \to K$. Hence there is a unique continuous map

$$\sigma : K \to W^{**}$$

such that

$$\sigma(\pi(h)) = q^{**}\nu_0(h) \qquad (h \in L).$$

The fiber calculation above also shows that σ is injective. Since K is compact and W^{**} is Hausdorff in the weak-star topology, σ is a homeomorphism of K onto the compact set $\sigma(K)$. The image $\sigma(K)$ is weak-star compact, hence weak-star closed and bounded, in W^{**}.

Thus K is homeomorphic to a weak-star closed subset of the second dual of the separable Rosenthal Banach space W. By Theorem 6.16 of [1], K is admissible. □

Theorem 2. *For a compact space K, the following are equivalent.*

1. *K is admissible.*
2. *K is homeomorphic to a compact subspace of a separable strongly Rosenthal compactum.*

Proof. If K is admissible, then $K \subset \mathrm{cls}_p(F)$ for a Rosenthal family $F \subset C(X)$ on a compact metrizable space X. Since $C(X)$ is separable, choose a countable norm-dense subset $D \subset F$. Then $\mathrm{cls}_p(D) = \mathrm{cls}_p(F)$: the inclusion $\mathrm{cls}_p(D) \subset \mathrm{cls}_p(F)$ is immediate, while every $f \in F$ is the uniform, hence pointwise, limit of a sequence from D. Hence $L = \mathrm{cls}_p(D)$ is a separable strongly Rosenthal compactum containing K.

Conversely, suppose that K is a compact subspace of a separable strongly Rosenthal compactum S. By Theorem 1, S is admissible. The class of admissible compacta is hereditary for compact subspaces, so K is admissible. □

Limitations

This settles the separable strongly Rosenthal case and the source paper's commented-out separable-envelope characterization. It does not settle the full source question because a strongly Rosenthal compactum need not be separable as a topological space, nor is it known here that every strongly Rosenthal compactum embeds into a separable strongly Rosenthal compactum. The proof of Theorem 1 uses a countable dense sequence in K in an essential way to build the metrizable graph compactification and countable Rosenthal family.

The full nonseparable question should therefore be read in the sharper form given by Theorem 2. A direct attack using all points of a nonseparable $K \subset \mathcal{B}_1(X)$ gives a graph compactification over $X \times [-1,1]^K$, but that space is generally nonmetrizable and the corresponding coordinate family is uncountable. The Banach-space quotient argument then naturally produces a representation over a nonseparable Rosenthal space, whereas admissibility requires a separable Rosenthal space. Thus the remaining obstruction is not the quotient step; it is the absence, in this pass, of a theorem producing a separable strongly Rosenthal envelope for an arbitrary strongly Rosenthal compactum.

Novelty and Search Notes

This packet is promoted as a partial result for the original question and as a full result for the separable subcase. The agent searched the local run indexes for arXiv:0803.2320, strongly Rosenthal, admissible compacta, and the previous cover packet. Targeted web searches for the phrases *separable strongly Rosenthal admissible compactum*, *every separable strongly Rosenthal admissible*, and close variants found no exact later statement in the bounded search. The source TeX itself contains a commented-out attempted implication for a subspace of a separable strongly Rosenthal compactum, stopping after the pointwise approximation step; the quotient argument above appears to fill the separable strongly Rosenthal case by routing through the admissible graph cover.

A final full-case literature/reduction pass checked nearby countable coding, analytic-subspace, and strong-boundedness machinery for separable Rosenthal compacta [5, 6, 7]. These tools are powerful precisely in the separable or analytic-family setting; they did not supply, in this pass, a separable envelope theorem for an arbitrary nonseparable strongly Rosenthal compactum.

Novelty should still be treated as tentative until a human checks nearby Rosenthal compacta folklore and the exact published version of the Glasner–Megrelishvili theorem.

Verification Notes

No computational verification is involved. The proof has three key review points:

- the graph-coordinate family (g_n) is Rosenthal because any finite-independence pattern in Y transfers back to the dense original graph $e(X)$;
- the source Theorem 6.3 representation is used with the trivial action and preserves the original coordinate functions as vectors of V;
- for $N = \bigcap_x \ker \alpha(e(x))$, the annihilator $N^\perp$ is exactly the closed linear span of the graph evaluations, so q^{**} identifies exactly the fibers of $\pi : L \to K$.

Human Review Recommendation

Treat this as a serious candidate proof of the separable version of the Glasner–Megrelishvili question. The most important line to audit is the fiber calculation for $q^{**} \circ \nu_0$. If that calculation is accepted, the rest is a direct application of the source paper's second-dual characterization of admissibility.

References

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